

Artificial Intelligence is revolutionizing healthcare through advanced diagnostic tools. Identify TWO ways AI can be used to analyze medical images for early disease detection.

Explain TWO functions of Artificial Intelligence in enabling autonomous decision-making in robots.

Outline FOUR reasons why a hospital management system should use the correct data types to capture patient data.

A warehouse robot is programmed to deliver items from shelves to packing stations based on specific delivery goals. Highlight FOUR steps, which the goal-based agent would follow to process environmental data to achieve this task.

Write a Python program that takes a user's full name as input, displays the name in all uppercase letters, counts and prints the total number of characters in the name.

In a payroll program, the variable $a = 5.5$ represents hours worked, and $b = 2.0$ represents the hourly rate. Write Python code that calculates and displays the total pay.

Company XT is developing smart systems to improve customer experience, and optimize decision-making using supervised machine learning technique. State TWO advantages of using this technique.

Distinguish between user-defined functions and built-in functions as used in Python.

The list below shows Python libraries. Give situations in which you will use each library.

- NumPy
- SciKit
- Pandas
- Matplotlib

A Computer Science trainee wants to install Python for use in machine learning. Outline FOUR steps the trainee should follow to complete the installation.

A company specializing in e-commerce has integrated AI applications into various aspects of their operations. Discuss FIVE impacts of AI applications on the company's operations.

Write a Python program that takes an integer input from the user representing a student's score and displays the competency of the student based on the following criteria:

Score	Competency
>=90	Excellent
Between 80 to 89	Good
Between 70 to 79	Average
Between 60 to 69	Pass
<60	Fail
Non-integer or negative score	Error message

An organization is developing an AI security system to detect and confirm potential intrusions like break-ins. Identify the type of inferencing that could be applied in this scenario.

Explain THREE reasons why this type of inferencing would help the AI system make accurate and reliable security decisions.

Artificial Intelligence (AI) is widely used in improving security systems. Discuss FOUR areas in which AI can be applied to prevent or detect fraudulent activities.

Explain TWO key characteristics of the K-Nearest Neighbors (KNN) algorithm in machine learning, focusing on its functionality.

Write a Python program using an if-else statement to determine if a number entered by the user is even or odd. Print "Even" if the number is even, and "Odd" if it is odd.

Logical operators are fundamental tools in problem solving. Using Python code example, illustrate the use of logical operators (AND, OR and NOT).

Describe the "naive" assumption in the Naive Bayes algorithm and provide an example to demonstrate how this assumption is applied.

Write a Python program that implements a function to calculate the area of a rectangle.

Explain FOUR ways in which inheritance in Object-Oriented Programming (OOP) helps in building complex and reusable code.

Describe FOUR approaches in which encapsulation helps in developing large-scale software systems, in managing system complexity and enhancing security.